

External Noise Interference

V2.0

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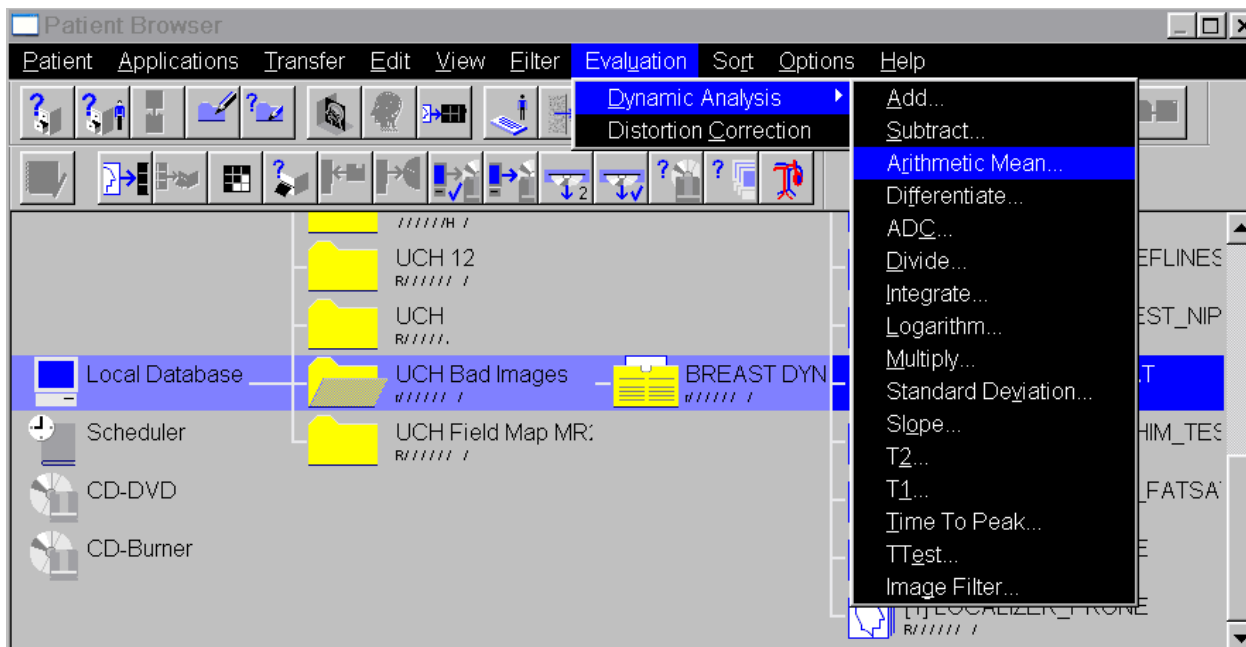
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How to identify if you have external interference

First step in identifying external noise is listening to the customer's image quality complaint, and then reviewing those images for obvious line artifacts. Sometimes those artifacts can be very subtle and hard to see, so performing a simple arithmetic mean of the entire series can be helpful in seeing the problem.

Example complaints can be lines, bands in the phase encoding direction, noisy, grainy, poor contrast. Let's start by performing an arithmetic mean of the image series customer is complaining about.

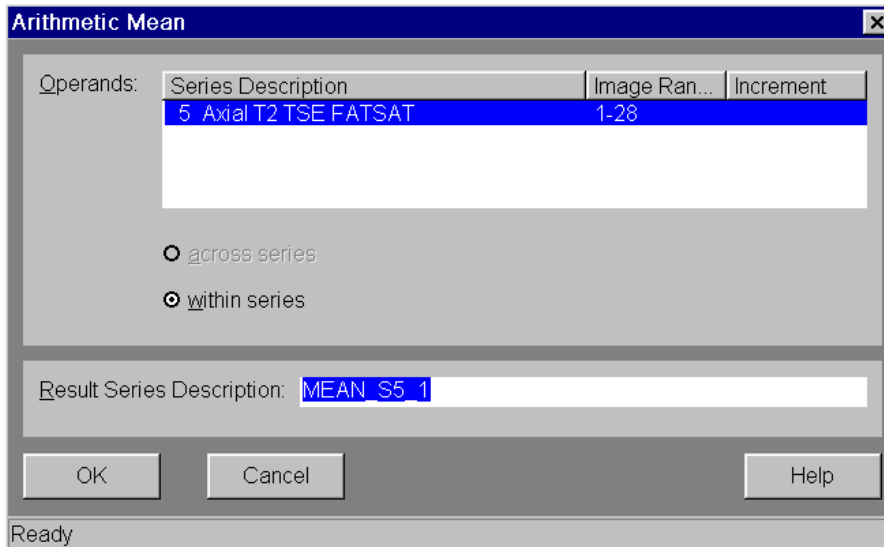
Open the patient browser and highlight the images series with the complaint. On top menu select Evaluation / Dynamic Analysis / Arithmetic Mean.



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On the next pop up window you have the option of giving the resultant image a meaningful name. Select OK then display the resultant image in the Viewing Tab



Window this image down to background noise levels and what was a subtle line that was barely noticeable are now very clear.



This technique is good for seeing noise bands, lines, dot patterns. Not all lines are external interference. You will need to window / center up to see if lines are caused by blood flow, CSI flow, etc.

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Some of the best tools from isolating external interference can be the customer protocols themselves. Try to see if problems can be seen in scout scans or other short duration scans. Then try to apply these scans using phantoms.

If you register the Service Patient under the Service platform / Test Tools / Tools, you can also place adjustments to off or service mode. This will enable you to perform scans without any adjustments being performed. Using this method you can disconnect the RF input to the RFPA run the scan and then look for lines, noise bands, etc without phantom brightness hiding them. This along with the arithmetic mean can expose external interference where none are usually seen.

Performing the RF_NOISE scan from the service sequences is a good test method but one has to understand how this sequence runs in order to understand the results properly.

The sequence will generate 50 images. The image covers +/- 250 KHz from current system frequency. System frequency will be placed at the crossing point between image 25 and 26. Only 255 columns are collected on each image resulting in each column representing about 39.22 hertz and each image covers 10 KHz, using this information one can calculate the frequency of artifacts.

Read the section entitled Procedures to Analyze the RF Noise Test and Procedure for Selecting RF Noise Test.

Once you have identified you have external interference then you can use the procedures for isolating for the specific modalities.

These procedures were written with RF noise sequence in mind but, can be adapted for customer protocols.

Important notes on procedures

1. If the patient table electronics is placed in virtual mode and not powered off this will cause an increase in noise seen by the scan since the electronics never receive any commands to shut down.
2. For Tim systems it is not possible to power off the MSUP but, this item still has potential to cause external interference.
3. For the Trio Whole Body the Pin Booster can potentially cause noise and/or spikes in sequences but, if disabled no signal can be measured.
4. The Rf_Noise sequence will normally perform an adjust frequency when it starts, for this reason it is imperative if no phantom is present that the tests are performed with the Service Patient registered and under Test Tools / Tools / Adjustment is set to either disabled or service. This will keep adjustments from being performed
5. In the procedures for running the RF noise sequence with the scan room equipment powered down, it is imperative that a scan is first performed with the system configured normally. This is needed to configure the components in the receive path before a loss of communication happens. Failure to do this can result in no signal during noise scan.
6. Items that can cause interference and are normally missed includes the following: Scan room lights, Energy efficient lights, Injectors, Charger units for injectors, In room monitors, Call buttons, Thermostats, Humidistat, Oxygen monitors, Power outlets, Wires passing thru waveguides, Damaged RF door seals and emergency lighting.
7. Usage of a cheap AM/FM radio can assist in determining there is a leak in the RF room. Use the radio outside the room, tune to a strong station in the FM band. Place in the RF room at various places especially around the filter plate. Reception in the room usually indicates a room leakage. An alternate would be to use a baby monitor and place the transmitter by the cold head then using the receiver outside the room to listen for signal, this has the advantage of being able to pin point the leak closer.

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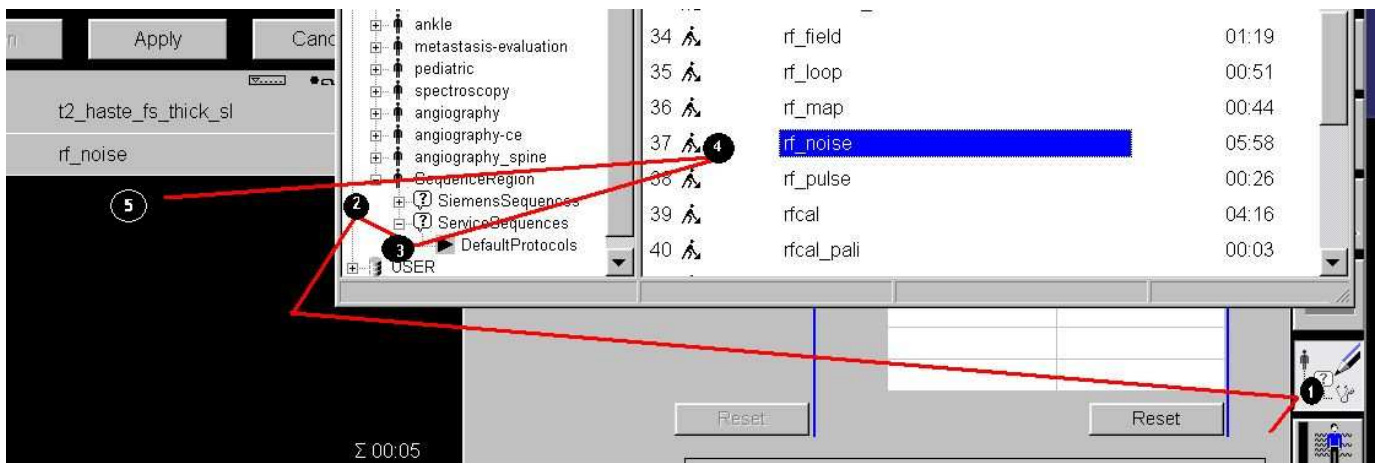
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Procedure for Selecting RF Noise Test

Procedures are meant for Body Coil but can be adapted for any coil if the Body coil is not sensitive enough ****Note**** SUM of SQUARES must be selected for any multi element coil or the resulting images may not represent the true noise level

1. Click on Exam Explorer
2. Click on + of "Sequence Region"
3. Click on + of "Service Sequence" then "Default Protocols"
4. Click on "rf_noise" and drag to area 5.
5. Sequence placed here.

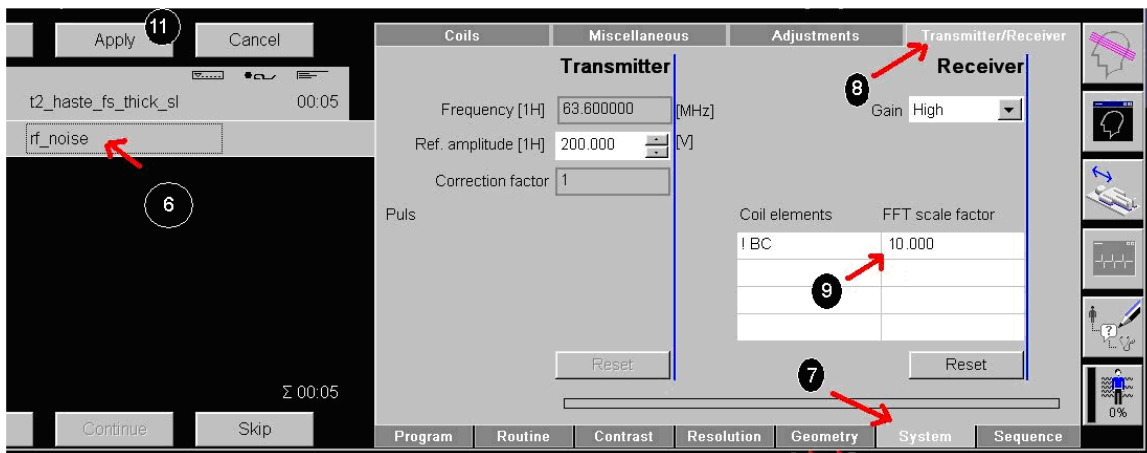


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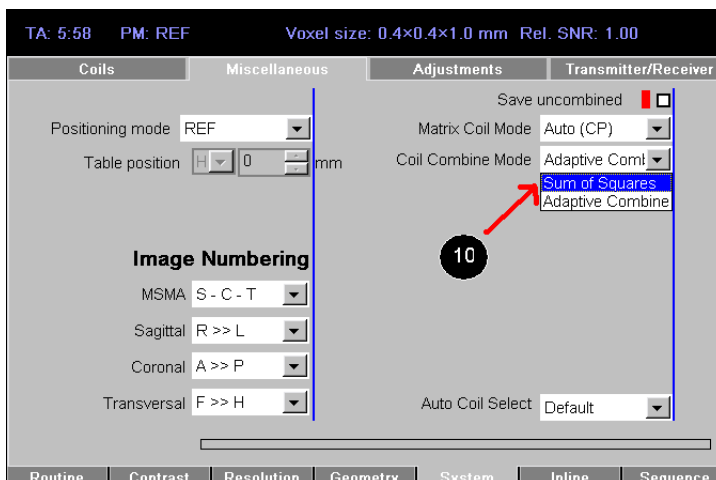
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6. Select sequence and select open.
7. Select "System Tab"
8. Select "Transmitter/Receiver Tab"
9. Change "FFT scale factor" to 10.0
10. If this is a Tim system and a multi element coil is used, select Sum of squares



11. Run sequence (Apply)



The purpose of setting the scale factor is to insure you do not have a data overflow in the "ADC" and allows direct comparison to other systems.

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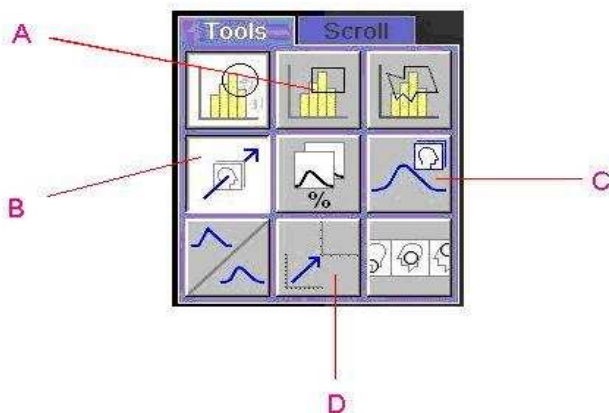
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Procedures to Analyze the RF Noise Test Using the Syngo Platform

The RF_noise sequence can be utilized to evaluate RFI noise as follows:

1. Evaluate resultant images with mean curve:
 - a. Viewing tab, Patient Browser, select 50 resultant images (Series Stack display)
 - b. Right-click noise image, click 'select series'
 - c. Click 'applications', click 'mean curve'
 - d. Select 'square' (from tools tab item A), and draw a large ROI square
 - e. Select 'within series' (from tools tab, selected by default item B), and click 'Start Evaluation' (from tools tab item C)
 - f. If you wish to have the graph display image numbers then select 'Scaling / Sorting' (From tools tab item D) then for axis label select 'Ser / Ima No'

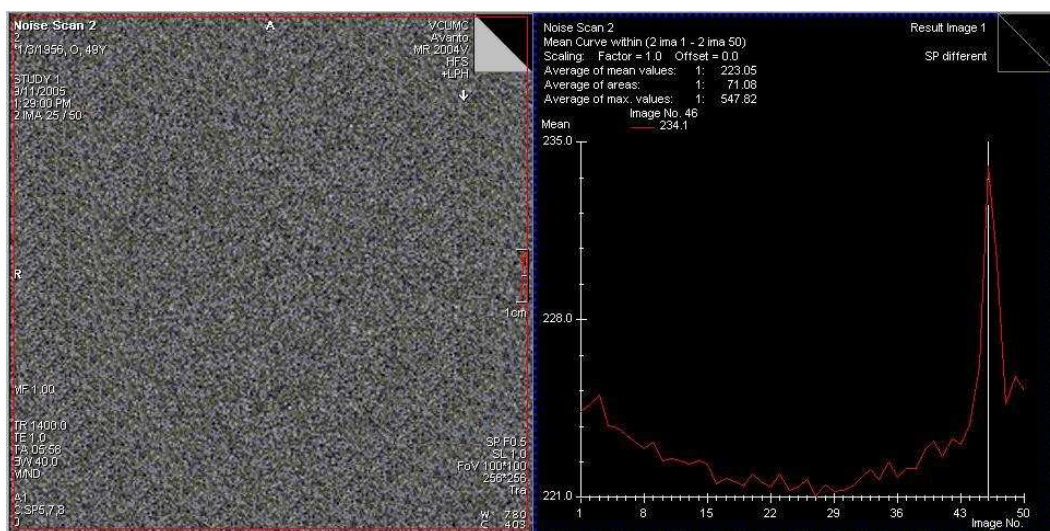


2. The graph displayed should show either a slope or curve. If any image has excessive noise or artifacts it can be seen as a spike or hump in the graph. If you had select the Image number for the axis display then the vertical intersect line can be moved and display what image this correspond to.

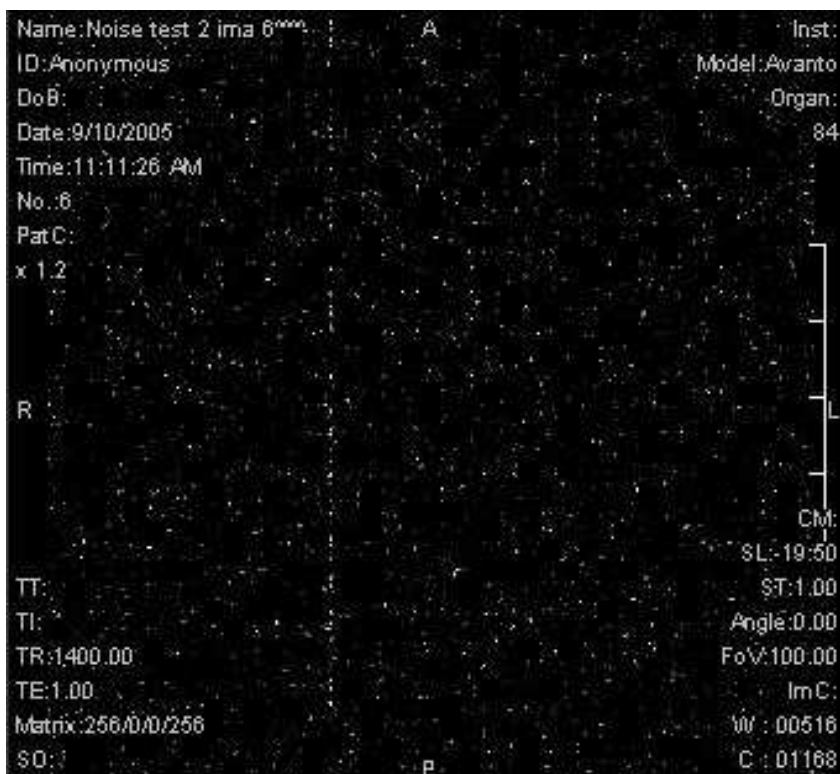
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- Each image should still be viewed for artifacts; a single line may not show up in the profile but can easily be seen in the image itself. Recommend to right click on first image and click on select series. Then adjust window / center for a clear view of background noise. This will make all images have the same window center.



4. If you wish to compare noise scans to other systems you have to compare same type systems as well as same coil configuration. To properly compare you must also set the scale factors of the two image sets to the same values.
5. First you need to get the scale factor of the images you are evaluating, do this by registering a patient, then Phoenix the image into the exam page, open the resulting sequence program, select the "System" tab, then on this page select the "Transmitter / Receiver" tab. Under the listed coil you will find the FFT scale that was used. Take this value and divide by 10 IE: (87.31 becomes 8.731)
6. Next, in the patient browser select the series containing the noise scan to modify, then select in the menu "Evaluation" then "Dynamic Analysis", then "Divide".
7. On this page under "Constant" enter the FFT scale divided by 10 you previously acquired. Optional, you can change the "Resultant series description" to a meaningful description.
8. Next select "OK" and a new series will be generated under the name supplied in "Resultant series description". You may need to refresh the browser.
9. This procedure will modify the images to a scale factor of 10 where they can be properly compared to other systems modified in the same way.

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Procedures to Analyze the RF Noise Test

Using the RFNOISE.COM Program

1. Extract the images either by doing an external export and copy the images to a floppy disk or USB drive. Or you can just archive the patient to CD.
2. The images can be analyzed by the RFNOISE.COM program directly off an archive CD by opening Windows Explorer and navigating to the CD drive and folder containing the images. Right click and use "SendTo" or drop and drag to the RFNOISE.COM program, depending on how you installed the program.

Noise Evaluation Program

Purpose:

This program is used to generate a report used to evaluate images created from usage of the Siemens RF_Noise sequence. It generates graphs representing the mean of each column of all 50 images. Each image is further broken down with graphs to allow more detail examination of the individual images along with the frequency for each column calculated.

Advantage of using this program over Syngo Mean Evaluation is this program evaluates each column data instead of each image as a whole. This allows more subtle artifacts or problems to be seen. The Syngo Mean Evaluation displays what could be described as passing thru a low pass filter not allowing small variations to be seen.

Restrictions:

This program usage is for Siemens use only. Program is not allowed to be run on a MR host system. Program checks for proper usage and will fail with an error message if an attempt to operate on a Syngo System.

Installation:

Installation is simple; two options are available depending how you would like to use the program. First option is to copy the file noise.com to the desktop of your personal computer. With this option, you will drop and drag one of the images from the group of noise images on top of the program. Second option would be to copy the file to the

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“SendTo” directory of your personal computer. This directory is located at “c:\Documents and Setting\login_name\SendTo”. Login_Name is the name you use to login into your personal computer. With this option you would select one image from the group of noise images, right click the mouse, select “Send To” and select the noise.com program. Under no circumstances is it allowed to copy this program on a Syngo system. The program will generate an error message indicating you are not authorized to run on that system.

Usage:

Depending on installation, drop and drag a single image on top of program, or select a single image, right click and send to this program. If the proper images are found then the program after report generation will launch your default web browser with the first page of graphs.

Noise Scan Evaluation

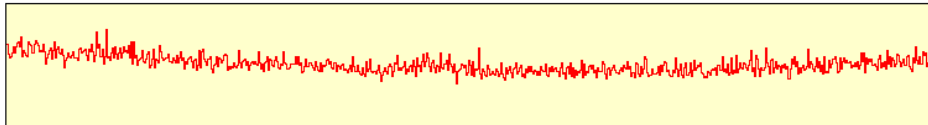
Version 2.0

SN 25707

2006-02-02 08:50

Frequency 63.395882 - 63.895882 Mhz

Center Frequency 63.645882 Mhz



Maximum Column Mean Value: 582

Minimum Column Mean Value: 425

[Top](#)

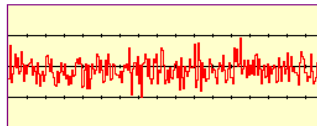


Image 01

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The top graph represents all 50 images. By placing the mouse cursor over an area of interest and clicking the left mouse button it will place you to the individual graph for that image.

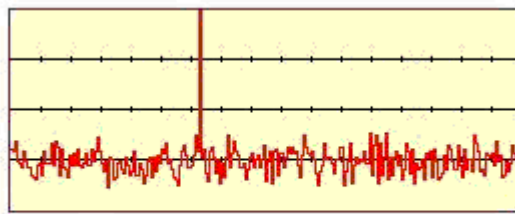


Image 06

If you click on this graph it will launch a new page with the mean values of all the column data belonging to this image along with a picture of the image.

Noise Evaluation

Image 06

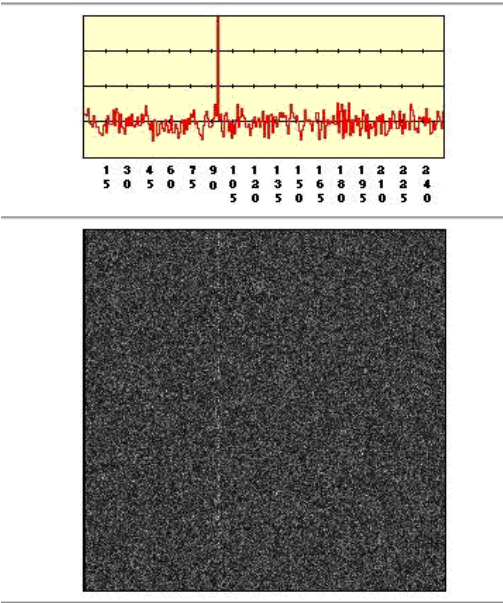


Image Data

Column Number	Mean Value
001	518
002	516
003	531

Additional Info:

Data generated is placed in directory “c:\temp\noise”. For those who would like to use the raw data from the noise scan the file “c:\temp\noise\ndata.txt” contains all data points (255 X 50) 12750.
The example file “Noise.xls” takes this data directly from c:\temp\noise and generates a broadband graph of the noise levels

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Procedures for Troubleshooting using the RF Noise Test

Harmony / Symphony

The following tests are meant as guidelines and do not constitute a complete checkout of the system that would completely exclude the Siemens equipment as the source of any problems. These tests make the assumption that you have found and confirmed you do have external noise problems.

1. Start by positioning the phantom in the magnet center. Acquire a short cable approximately 7 feet in length. A suitable item would be a telephone cord but, any wire without magnetic parts will work. Place this wire starting at the center of the magnet and stretch out on the patient table extending out the front and rear of the magnet. This simulates a person's body and allows for testing in a like condition with a patient.
2. Open service platform, Select "Test Tools" , "Tools" , "Service Patient Registered" and "Adjustments disabled / Adjustments service"
3. Open Exam Explorer find the rf_noise sequence under Siemens Service Sequences, then drop drag sequence into measurement area.
4. Open sequence then on the "System" tab, select the coil you wish to use for the test and under "Transmitter / Receiver" tab change FFT scale to 10.0. (See "Procedure for Selecting RF Noise Test"). Do not select Apply, Instead to perform scan always select the scan button.
5. The sequence will generate 50 images. The image covers +/- 250 KHz from current system frequency. System frequency will be placed at the crossing point between image 25 and 26. Only 255 columns are collected on each image resulting in each column representing about 39.22 hertz and each image covers 10 KHz, using this information one can calculate the frequency of artifacts. Use the following section "Procedure to Analyze the RF Noise Test" on how to examine these images.
6. Perform scan by selecting "Scan" and verify you have artifacts. If you do not have any artifacts in the images you might try using a more sensitive coil such as an extremity coil. The remainder of these tests assumes you have artifacts in the images at this point.
7. Steps 8 thru 10 can be performed together to save time.

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8. Remove or power off any customer equipment located in scan room, this includes any battery chargers and anything that runs on batteries and re-run scan.
9. Remove power to the Siemens intercom unit. Re-run scan and see if the artifacts go away.
10. Switch off Cold Head Compressor and Re-scan.
11. If artifacts are still present then place gradient cabinet out of supervision and into virtual mode, then power down gradient cabinet this includes GPA and main power to cabinet and re-run scan. If artifacts are not present then perform step 12 to determine if noise is being introduced in the scan room or cross talk inside the cabinets, start troubleshooting gradient cabinet / cables.
12. If artifacts are present at this point then have the body coil selected as the receiving coil. Unplug all coils and remove from table. Coil removal is required because cross talk on surface coil cables can interfere with this test. Terminate the output of the BTB with a 50 ohm termination. Perform the noise scan
13. Restore cable connections to BTB.
14. If noise is present then problem is internal with the measurement section. Start by isolating LCCS, RFCL, Receiver module, RX4
15. If advance shim is present then place MSU out of supervision and into virtual mode, switch off circuit breaker and re-run scan.
16. If artifacts are still present then go into “Test Tools / Supervision” Then set PTAB, BTB1, BTB2, and RFAS out of supervision and into Virtual Mode).
17. Switch off Circuit breaker F22 or F15 and kill power to scan room equipment. Power for the measurement section comes from a different source and will not be affected. Re-run scan and check for artifacts. If artifacts disappear you may have to reboot scanner and host to get artifacts to return after you restore power. This reboot of scanner and host is required to get system back in normal operating mode.
18. If the artifacts are gone then isolation of the individual components that was powered off are required
19. If artifacts are still present then the cause is most likely external to the Siemens equipment. Check the filter plate for improperly configured connections, cables passing thru waveguides. Check for room thermostats, intercoms, etc. Any external connection needs to be removed to verify that it is not the cause of the problem. Check obvious items such as the RF door, any RF seals. If nothing can be found then an RF integrity test may be required from an external vendor.

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20. Further tests can be performed by comparing the intensity of noise lines with the system configured normal and opening the scan room door and perform a scan. This would not conclude the door is a problem but may give you a direction the noise is coming from. You can perform the same test with a wire placed thru the filter plate wave guide. If in either case the individual noise line intensifies then that could be the direction of the noise source. **Note:** This test can be misleading. You must compare the intensity of the original noise line with the same noise line of the new scan. Expect to have addition lines with the wire or door, only compare the original lines.

Typical values for noise scan on the Symphony:

Body coil: High mean value of 210, low mean value 100
BTB terminated: Mean value 95
BCCS terminated: Mean value 145
LCCS terminated: Mean value 12
Receiver terminated: Mean value 5

Procedures for Troubleshooting using the RF Noise Test

Trio Whole Body

The following tests are meant as guidelines and do not constitute a complete checkout of the system that would completely exclude the Siemens equipment as the source of any problems. These tests make the assumption that you have found and confirmed you do have external noise problems.

1. Start by positioning the phantom in the magnet center. Acquire a short cable approximately 7 feet in length. A suitable item would be a telephone cord but, any wire without magnetic parts will work. Place this wire starting at the center of the magnet and stretch out on the patient table extending out the front and rear of the magnet. This simulates a person's body and allows for testing in a like condition with a patient.
2. Open service platform, Select "Test Tools" , "Tools" , "Service Patient Registered" and "Adjustments disabled / Adjustments service"
3. Open Exam Explorer find the rf_noise sequence under Siemens Service Sequences, then drop drag sequence into measurement area.
4. Open sequence then on the "System" tab, select the coil you wish to use for the test and under "Transmitter / Receiver" tab change FFT scale to 10.0. (See "Procedure for Selecting RF Noise Test"). Do not select Apply, Instead to perform scan always select the scan button.
5. The sequence will generate 50 images. The image covers +/- 250 KHz from current system frequency. System frequency will be placed at the crossing point between image 25 and 26. Only 255 columns are collected on each image resulting in each column representing about 39.22 hertz and each image covers 10 KHz, using this information one can calculate the frequency of artifacts. Use the following section "Procedure to Analyze the RF Noise Test" on how to examine these images.
6. Perform scan by selecting "Scan" and verify you have artifacts. If you do not have any artifacts in the images you might try using a more sensitive coil such as an extremity coil. The remainder of these tests assumes you have artifacts in the images at this point.
7. Steps 8 thru 10 can be performed together to save time.

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8. Remove or power off any customer equipment located in scan room, this includes any battery chargers and anything that runs on batteries and re-run scan.
9. Remove power to the Siemens intercom unit. Re-run scan and see if the artifacts go away.
10. Switch off Cold Head Compressor and Re-scan
11. If artifacts are still present then place gradient cabinet out of supervision and into virtual mode, then power down gradient cabinet this includes GPA and main power to cabinet and re-run scan. If artifacts are not present then perform step 12 to determine if noise is being introduced in the scan room or cross talk inside the cabinets, then start troubleshooting gradient cabinet / cables.
12. If artifacts are present at this point then have the body coil selected as the receiving coil. Unplug all coils and remove from table. Coil removal is required because cross talk on surface coil cables can interfere with this test. Terminate the output of the BTB with a 50 ohm termination. Perform the noise scan
13. Restore cable connections to BTB.
14. If noise is present then problem is internal with the measurement section. Start by isolating LCCS, RFCI, Receiver module, RX4
15. If advance shim is present then place advance shim out of supervision and into virtual mode, switch of circuit breaker and re-run scan.
16. If artifacts are still present then go into “Test Tools / Supervision” Then set PTAB, BTB1, BTB2, and RFAS out of supervision and into Virtual Mode).
17. Disconnect connector X100 on patient table control rack to kill power to scan room equipment. Disconnection here is required since Pin booster gets power from same source. Power for the measurement section comes from a different source and will not be affected. Re-run scan and check for artifacts. If artifacts disappear you may have to reboot scanner and host to get artifacts to return after you restore.
18. If the artifacts are gone then isolation of the individual components that was powered off are required
19. If artifacts are still present then the cause is most likely external to the Siemens equipment. Check the filter plate for improperly configured connections, cables passing thru waveguides. Check for room thermostats, intercoms, etc. Any external connection needs to be removed to verify that it is not the cause of the problem. Check obvious items such as the RF door, any RF seals. If nothing can be found then an RF integrity test may be required from an external vendor.

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20. Further tests can be performed by comparing the intensity of noise lines with the system configured normal and opening the scan room door and perform a scan. This would not conclude the door is a problem but may give you a direction the noise is coming from. You can perform the same test with a wire placed thru the filter plate wave guide. If in either case the individual noise line intensifies then that could be the direction of the noise source. **Note:** This test can be misleading. You must compare the intensity of the original noise line with the same noise line of the new scan. Expect to have addition lines with the wire or door, only compare the original lines.

Procedures for Troubleshooting using the RF Noise Test

Concerto / C!

The following tests are meant as guidelines and do not constitute a complete checkout of the system that would completely exclude the Siemens equipment as the source of any problems. These tests make the assumption that you have found and confirmed you do have external noise problems.

1. Start by positioning the phantom in the magnet center with the Head or Body Spine coil. Acquire a short cable approximately 7 feet in length. A suitable item would be a telephone cord but, any wire without magnetic parts will work. Place this wire starting at the center of the magnet and stretch out on the patient table extending out the right and left of the magnet. This simulates a person's body and allows for testing in a like condition with a patient.
2. Open service platform, Select "Test Tools" , "Tools" , "Service Patient Registered" and "Adjustments disabled / Adjustments service"
3. Open Exam Explorer find the RF noise sequence under Siemens Service Sequences, then drop drag sequence into measurement area.
4. Open sequence then on the "System" tab, select the coil you wish to use for the test and under "Transmitter / Receiver" tab change FFT scale to 10.0. (See "Procedure for Selecting RF Noise Test"). Do not select Apply, Instead to perform scan always select the scan button.
5. The sequence will generate 50 images. The image covers +/- 250 KHz from current system frequency. System frequency will be placed at the crossing point between image 25 and 26. Only 255 columns are collected on each image resulting in each column representing about 39.22 hertz and each image covers 10 KHz, using this information one can calculate the frequency of artifacts. Use the following section "Procedure to Analyze the RF Noise Test" on how to examine these images.
6. Perform scan by selecting "Scan" and verify you have artifacts. The remainder of these tests assumes you have artifacts in the images at this point.
7. Remove or power off any customer equipment located in scan room, this includes any battery chargers and anything that runs on batteries and re-run scan.
8. If artifacts are still present then place gradient cabinet out of supervision and into virtual mode, then power down gradient cabinet this includes GPA and main power to cabinet and re-run scan.

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If artifacts are not present then perform step 10 to determine if noise is being introduced in the scan room or cross talk inside the cabinets, then start troubleshooting gradient cabinet / cables.

9. If artifacts are present at this point then place the TCU out of supervision and in virtual mode and re-run scan
10. If artifacts present terminate input to Cosimux X1/X2 (Plug 1) and/or X5/X6 (Plug 2) based on plug used and coil configuration and rerun scan.
11. If noise is present then problem is internal with the measurement section. Start by isolating Cosimux, RFCI, Receiver module, RX4
12. If not present, check the filter plate for improperly configured connections, cables passing thru waveguides. Check for room thermostats, intercoms, etc. Any external connection needs to be removed to verify that it is not the cause of the problem. Check obvious items such as the RF door, any RF seals. If nothing can be found then an RF integrity test may be required from an external vendor.
13. Further tests can be performed by comparing the intensity of noise lines with the system configured normal and opening the scan room door and perform a scan. This would not conclude the door is a problem but may give you a direction the noise is coming from. You can perform the same test with a wire placed thru the filter plate wave guide. If in either case the individual noise line intensifies then that could be the direction of the noise source. **Note:** This test can be misleading. You must compare the intensity of the original noise line with the same noise line of the new scan. Expect to have addition lines with the wire or door, only compare the original lines.

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Procedures for Troubleshooting using the RF Noise Test

Avanto / Espree under VB13 software

The following tests are meant as guidelines and do not constitute a complete checkout of the system that would completely exclude the Siemens equipment as the source of any problems. These tests make the assumption that you have found and confirmed you do have external noise problems.

1. Start by positioning the phantom in the magnet center. Acquire a short cable approximately 7 feet in length. A suitable item would be a telephone cord but, any wire without magnetic parts will work. Place this wire starting at the center of the magnet and stretch out on the patient table extending out the front and rear of the magnet. This simulates a person's body and allows for testing in a like condition with a patient.
2. Open service platform, Select "Test Tools", "Tools", "Service Patient Registered" and "Adjustments disabled / Adjustments service"
3. Open Exam Explorer find the rf_noise sequence under Siemens Service Sequences, then drop drag sequence into measurement area.
4. Open sequence then on the "System" tab, select the coil you wish to use for the test and under "Transmitter / Receiver" tab change FFT scale to 10.0. (See "Procedure for Selecting RF Noise Test"). Do not select Apply, Instead to perform scan always select the scan button.
5. The sequence will generate 50 images. The image covers +/- 250 KHz from current system frequency. System frequency will be placed at the crossing point between image 25 and 26. Only 255 columns are collected on each image resulting in each column representing about 39.22 hertz and each image covers 10 KHz, using this information one can calculate the frequency of artifacts. Use the following section "Procedure to Analyze the RF Noise Test" on how to examine these images.
6. Perform scan by selecting "Scan" and verify you have artifacts. If you do not have any artifacts in the images you might try using a more sensitive coil such as an extremity coil. If a multi element coil is used and this is a Tim system turn **off** Adaptive Coil Combine and turn **on** Sum of Squares. This can be located on System/Miscellaneous tab. The remainder of these tests assumes you have artifacts in the images at this point.
7. Steps 8 thru 10 can be performed together to save time.

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8. Remove or power off any customer equipment located in scan room, this includes any battery chargers and anything that runs on batteries and re-run scan.
9. Remove power to the Siemens intercom unit. Re-run scan and see if the artifacts go away.
10. Switch off Cold Head Compressor and Re-scan
11. If artifacts are still present then place gradient cabinet out of supervision and into virtual mode, then power down gradient cabinet this includes GPA and main power to cabinet and re-run scan. If artifacts are not present then perform step 12 to determine if noise is being introduced in the scan room or cross talk inside the cabinets, then start troubleshooting gradient cabinet / cables.
12. If artifacts are present at this point then have the body coil selected as the receiving coil. Unplug all coils and remove from table. Coil removal is required because cross talk on surface coil cables can interfere with this test. Terminate the output of the Tales with a 50 ohm termination and perform the noise scan
13. Restore cable connections to Tales.
14. If noise is present then problem is internal with the measurement section. Start by RCCS, /RFIS, Receiver module.
15. If advance shim is present then place advance shim out of supervision and into virtual mode, switch of circuit breaker and re-run scan.
16. If artifacts are still present then go into “Test Tools / Supervision” Then set PTAB, RFIS, and RFAS out of supervision and Virtual Mode on.
17. Switch off Circuit breaker F22 and kill power to scan room equipment. Power for the measurement section comes from a different source and will not be affected. Re-run scan and check for artifacts. If artifacts disappear you may have to reboot scanner and host to get artifacts to return after you restore power. This reboot of scanner and host is required to get system back in normal operating mode.
18. If the artifacts are gone then isolation of the individual components that was powered off are required. **Note: For Tim systems a complete reboot of scanner and host is required for system to operate properly at this point. Failure to do this will result in additional noise in system not related to original problem**
19. If artifacts are still present then the cause is most likely external to the Siemens equipment. Check the filter plate for improperly configured connections, cables passing thru waveguides. Check for room thermostats, intercoms, etc. Any external connection needs to be removed to verify that it is

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not the cause of the problem. Check obvious items such as the RF door, any RF seals. If nothing can be found then an RF integrity test may be required from an external vendor.

20. Further tests can be performed by comparing the intensity of noise lines with the system configured normal and opening the scan room door and perform a scan. This would not conclude the door is a problem but may give you a direction the noise is coming from. You can perform the same test with a wire placed thru the filter plate wave guide. If in either case the individual noise line intensifies then that could be the direction of the noise source. **Note:** This test can be misleading. You must compare the intensity of the original noise line with the same noise line of the new scan. Expect to have addition lines with the wire or door, only compare the original lines.

Notes:

These procedures do not eliminate the MSUP in the Tim system

Procedures for Troubleshooting using the RF Noise Test

ESSENZA VC11

The following tests are meant as guidelines and do not constitute a complete checkout of the system that would completely exclude the Siemens equipment as the source of any problems. These tests make the assumption that you have found and confirmed you do have external noise problems.

1. Start by positioning the phantom in the magnet center. Acquire a short cable approximately 7 feet in length. A suitable item would be a telephone cord but, any wire without magnetic parts will work. Place this wire starting at the center of the magnet and stretch out on the patient table extending out the front and rear of the magnet. This simulates a person's body and allows for testing in a like condition with a patient.
2. Open service platform, Select "Test Tools" , "Tools" , "Service Patient Registered" and "Adjustments disabled / Adjustments service"
3. Open Exam Explorer find the RF_Noise sequence under Siemens Service Sequences, then drop drag sequence into measurement area.
4. Open sequence then on the "System" tab, select the coil you wish to use for the test and under "Transmitter / Receiver" tab change FFT scale to 10.0. (See "Procedure for Selecting RF Noise Test"). Do not select Apply, Instead to perform scan always select the scan button.
5. The sequence will generate 50 images. The image covers +/- 250 KHz from current system frequency. System frequency will be placed at the crossing point between image 25 and 26. Only 255 columns are collected on each image resulting in each column representing about 39.22 hertz and each image covers 10 KHz, using this information one can calculate the frequency of artifacts. Use the following section "Procedure to Analyze the RF Noise Test" on how to examine these images.
6. Perform scan by selecting "Scan" and verify you have artifacts. If you do not have any artifacts in the images you might try using a more sensitive coil such as an extremity coil. If a multi element coil is used and this is a Tim system turn **off** Adaptive Coil Combine and turn **on** Sum of Squares. This can be located on System/Miscellaneous tab. The remainder of these tests assumes you have artifacts in the images at this point.
7. Steps 8 thru 10 can be performed together to save time.

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8. Remove or power off any customer equipment located in scan room, this includes any battery chargers and anything that runs on batteries and re-run scan.
9. Remove power to the Siemens intercom unit. Re-run scan and see if the artifacts go away.
10. Switch off Cold Head Compressor and Re-scan
11. If artifacts are still present then place gradient cabinet out of supervision and into virtual mode, then power down gradient cabinet main power to amplifiers (F1101 on cabinet top) leaving power to the GPA off and re-run scan. If artifacts are not present then perform step 12 to determine if noise is being introduced in the scan room or cross talk inside the cabinets, then start troubleshooting gradient cabinet / cables. ****NOTE**** The ESSENZA will fault out if GPA is powered off no matter what the status of supervision is. *****WARNING***** Before you restore power with breaker F101 you must power off F26 to the GSSU first or power entire system down. **Failure to do this can cause extreme component damage**
12. If artifacts are present at this point then have the body coil selected as the receiving coil. Unplug all coils and remove from table. Coil removal is required because cross talk on surface coil cables can interfere with this test. Terminate the output of the Tales with a 50 ohm termination and perform the noise scan
13. Restore cable connections to Tales.
14. If noise is present then problem is internal with the measurement section. Start by RCCS, /RFIS, Receiver module.
15. If advance shim is present then place advance shim out of supervision and into virtual mode, switch of circuit breaker and re-run scan.
16. If artifacts are still present then go into “Test Tools / Supervision” Then set PTAB, RFIS, and RFAS out of supervision and Virtual Mode on.
17. Switch off Circuit breaker F22 and kill power to scan room equipment. Power for the measurement section comes from a different source and will not be affected. Re-run scan and check for artifacts. If artifacts disappear you may have to reboot scanner and host to get artifacts to return after you restore power. This reboot of scanner and host is required to get system back in normal operating mode.

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18. If the artifacts are gone then isolation of the individual components that was powered off are required. **Note: For the ESSENZA a complete reboot of scanner and host is required for system to operate properly at this point. Failure to do this will result in additional noise in system not related to original problem**
19. If artifacts are still present then the cause is most likely external to the Siemens equipment. Check the filter plate for improperly configured connections, cables passing thru waveguides. Check for room thermostats, intercoms, etc. Any external connection needs to be removed to verify that it is not the cause of the problem. Check obvious items such as the RF door, any RF seals. If nothing can be found then an RF integrity test may be required from an external vendor.
20. Further tests can be performed by comparing the intensity of noise lines with the system configured normal and opening the scan room door and perform a scan. This would not conclude the door is a problem but may give you a direction the noise is coming from. You can perform the same test with a wire placed thru the filter plate wave guide. If in either case the individual noise line intensifies then that could be the direction of the noise source. **Note:** This test can be misleading. You must compare the intensity of the original noise line with the same noise line of the new scan. Expect to have addition lines with the wire or door, only compare the original lines.

Notes:

These procedures do not eliminate the MSUP.

Procedures for Troubleshooting using the RF Noise Test

Tim systems VB13 and higher software

The following tests are meant as guidelines and do not constitute a complete checkout of the system that would completely exclude the Siemens equipment as the source of any problems. These tests make the assumption that you have found and confirmed you do have external noise problems.

1. Start by positioning the phantom in the magnet center. Acquire a short cable approximately 7 feet in length. A suitable item would be a telephone cord but, any wire without magnetic parts will work. Place this wire starting at the center of the magnet and stretch out on the patient table extending out the front and rear of the magnet. This simulates a person's body and allows for testing in a like condition with a patient.
2. Open service platform, Select "Test Tools", "Tools", "Service Patient Registered" and "Adjustments disabled / Adjustments service"
3. Open Exam Explorer find the rf_noise sequence under Siemens Service Sequences, then drop drag sequence into measurement area.
4. Open sequence then on the "System" tab, select the coil you wish to use for the test and under "Transmitter / Receiver" tab change FFT scale to 10.0. (See "Procedure for Selecting RF Noise Test"). Do not select Apply, Instead to perform scan always select the scan button.
5. The sequence will generate 50 images. The image covers +/- 250 KHz from current system frequency. System frequency will be placed at the crossing point between image 25 and 26. Only 255 columns are collected on each image resulting in each column representing about 39.22 hertz and each image covers 10 KHz, using this information one can calculate the frequency of artifacts. Use the following section "Procedure to Analyze the RF Noise Test" on how to examine these images.
6. Perform scan by selecting "Scan" and verify you have artifacts. If you do not have any artifacts in the images you might try using a more sensitive coil such as an extremity coil. If a multi element coil is used and this is a Tim system turn **off** Adaptive Coil Combine and turn **on** Sum of Squares. This can be located on System/Miscellaneous tab. The remainder of these tests assumes you have artifacts in the images at this point.
7. Steps 8 thru 10 can be performed together to save time.

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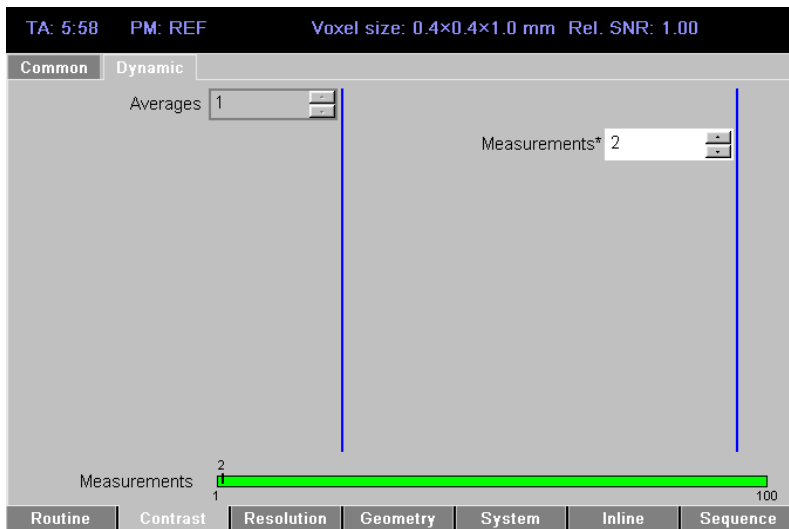
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8. Remove or power off any customer equipment located in scan room, this includes any battery chargers and anything that runs on batteries and re-run scan.
9. Remove power to the Siemens intercom unit. Re-run scan and see if the artifacts go away.
10. Switch off Cold Head Compressor and Re-scan
11. If artifacts are still present then place gradient cabinet out of supervision and into virtual mode, then power down gradient cabinet this includes GPA and main power to cabinet and re-run scan. If artifacts are not present then perform step 12 to determine if noise is being introduced in the scan room or cross talk inside the cabinets, then start troubleshooting gradient cabinet / cables.
12. If artifacts are present at this point then have the body coil selected as the receiving coil. Unplug all coils and remove from table. Coil removal is required because cross talk on surface coil cables can interfere with this test. Terminate the output of the Tales with a 50 ohm termination and perform the noise scan
13. Restore cable connections to Tales.
14. If noise is present then problem is internal with the measurement section. Start by isolating RCCS, RFIS, and Receiver module.
15. If advance shim is present then place advance shim out of supervision and into virtual mode, switch of circuit breaker and re-run scan.
16. If artifacts are still present then go into “Test Tools / Supervision” Then set PTAB, RFIS, and RFAS out of supervision and Virtual Mode on.
17. On measurement card select Contrast / Dynamic tab, here select Measurements to 2.

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18. VB13 and higher Software will not allow the scan to start with equipment room power turned off but will allow the second measurement to continue without any trouble. Run scan at this time. After scan starts turn off breaker F22. When both scans are completed discard first measurement and review second measurement. If artifacts disappear you may have to reboot scanner and host to get artifacts to return after you restore power. This reboot of scanner and host is required to get system back in normal operating mode.
19. If the artifacts are gone then isolation of the individual components that was powered off are required. **Note: For Tim systems a complete reboot of scanner and host is required for system to operate properly at this point. Failure to do this will result in additional noise in system not related to original problem**
20. If artifacts are still present then the cause is most likely external to the Siemens equipment. Check the filter plate for improperly configured connections, cables passing thru waveguides. Check for room thermostats, intercoms, etc. Any external connection needs to be removed to verify that it is not the cause of the problem. Check obvious items such as the RF door, any RF seals. If nothing can be found then an RF integrity test may be required from an external vendor.
21. Further tests can be performed by comparing the intensity of noise lines with the system configured normal and opening the scan room door and perform a scan. This would not conclude the door is a problem but may give you a direction the noise is coming from. You can perform the same test with a wire placed thru the filter plate wave guide. If in either case the individual noise line intensifies then that could be the direction of the noise source. **Note:** This test can be misleading. You must compare the intensity of the original noise line with the same noise line of the new scan. Expect to have addition lines with the wire or door, only compare the original lines.

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Notes:

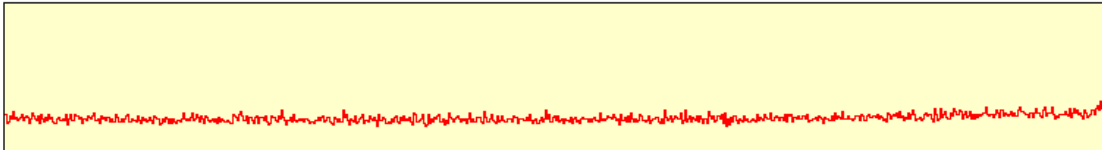
These procedures do not eliminate the MSUP in the Tim system

Typical Noise Scans

Avanto: Body Coil with Scale Factor = 10

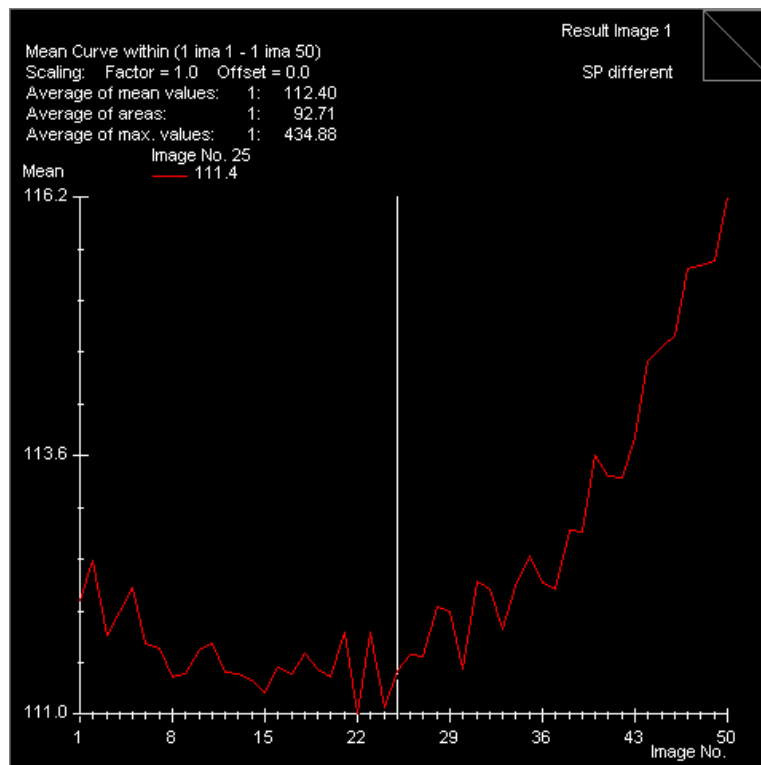
Frequency 63.430415 - 63.930415 Mhz

Center Frequency 63.680415 Mhz



Maximum Column Mean Value: 130

Minimum Column Mean Value: 97



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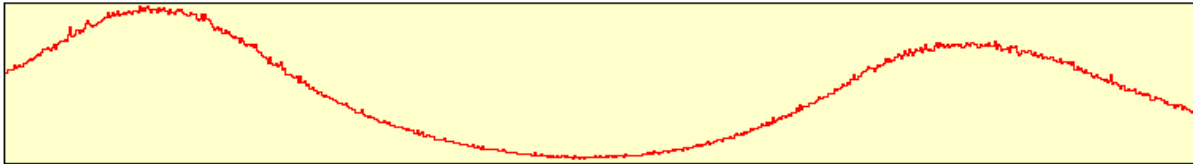
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Concerto: Body Spine Coil with Scale Factor = 10 Body Spine Coil

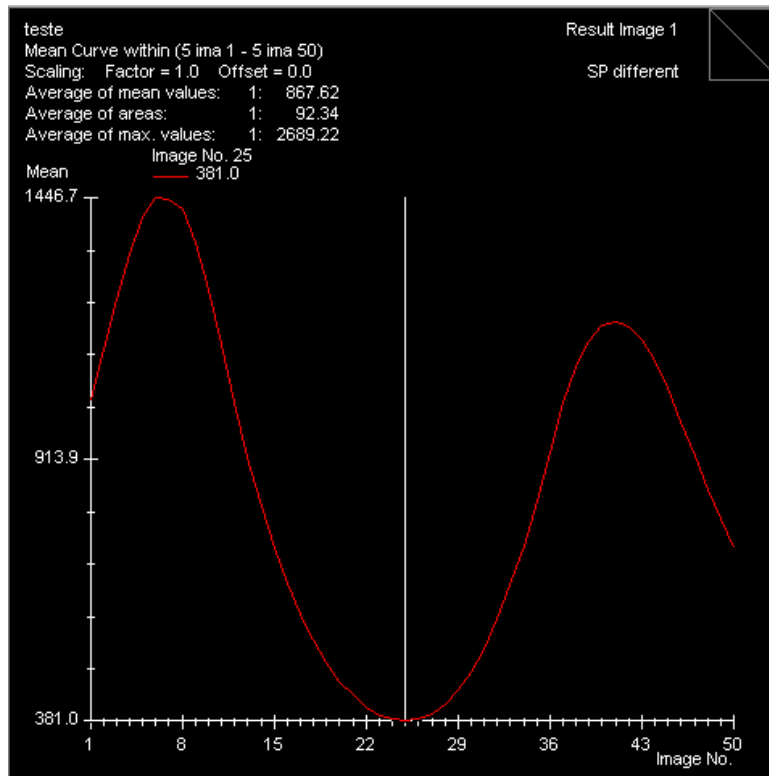
Frequency 8.0277641 - 8.5277641 Mhz

Center Frequency 8.277764 Mhz



Maximum Column Mean Value: 1538

Minimum Column Mean Value: 355

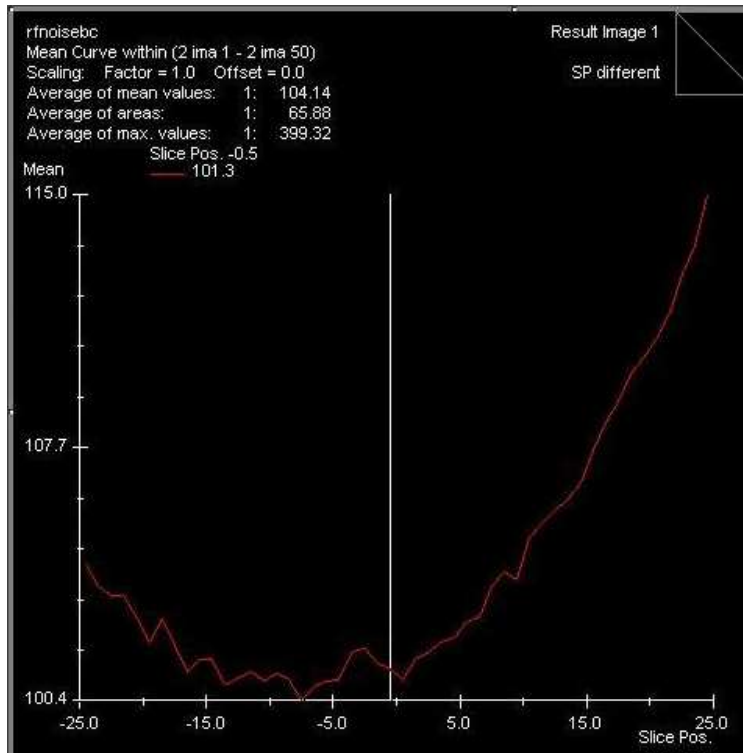


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Espreo Slim line: Body Coil with Scale Factor = 10



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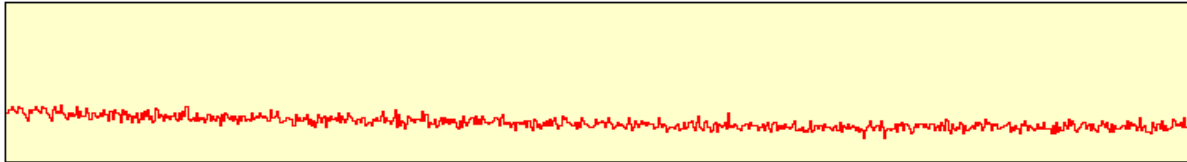
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ESSENZA: Body Coil with Scale Factor = 10

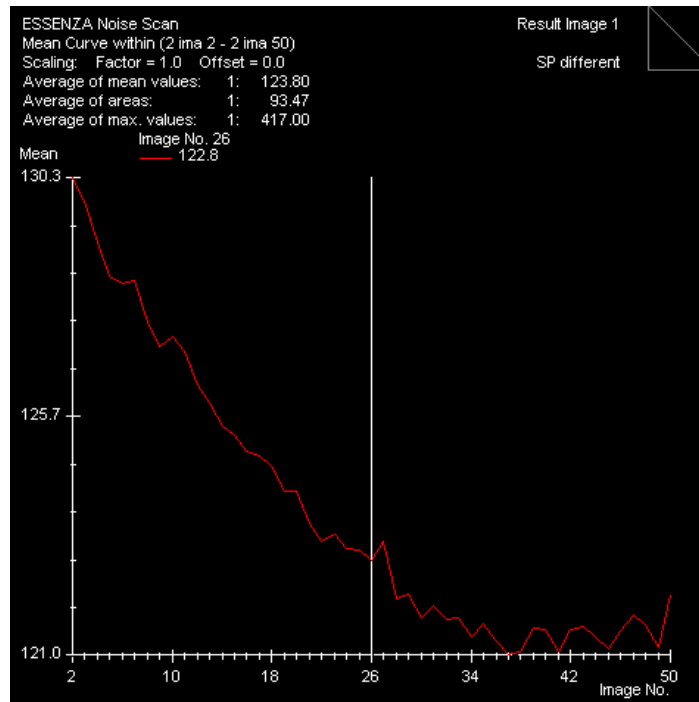
Frequency 63.399383 - 63.899383 Mhz

Center Frequency 63.649383 Mhz



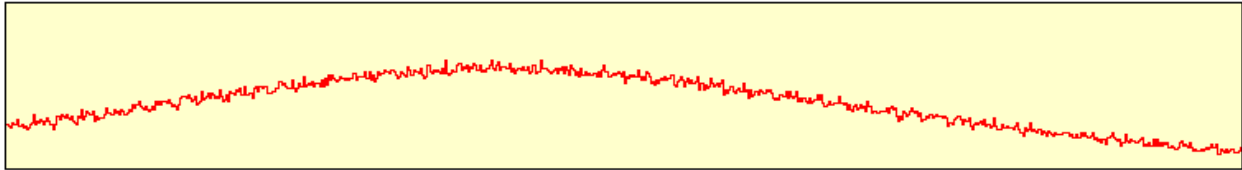
Maximum Column Mean Value: 142

Minimum Column Mean Value: 107



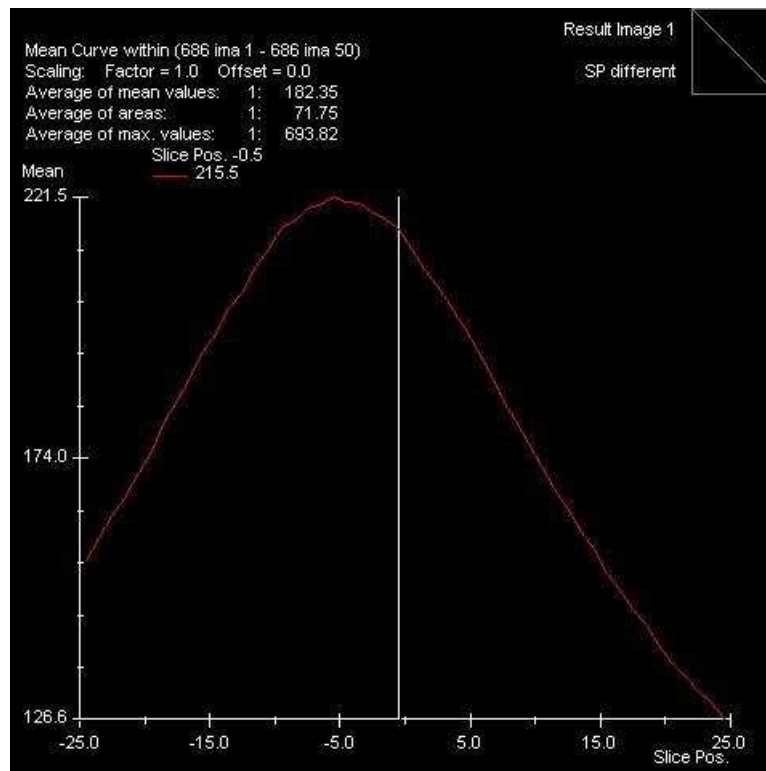
Harmony: Body Coil with Scale Factor = 10

Frequency 40.285167 - 40.785167 Mhz



Maximum Column Mean Value: 244

Minimum Column Mean Value: 114



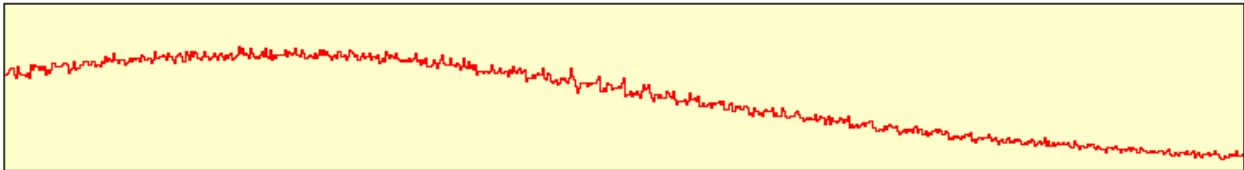
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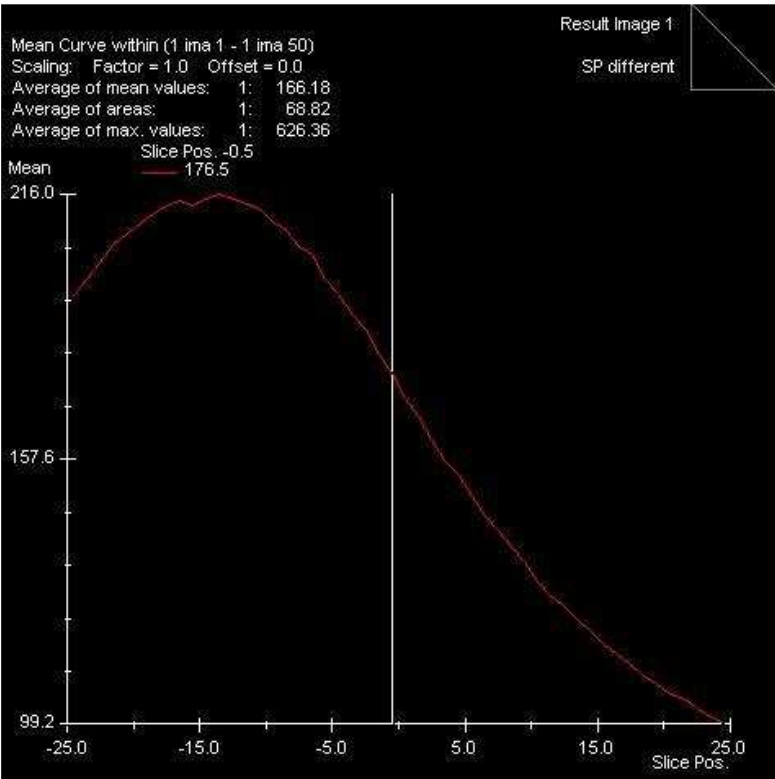
Symphony: Body Coil with Scale Factor = 10

Frequency 63.425665 - 63.925665 Mhz



Maximum Column Mean Value: 236

Minimum Column Mean Value: 89



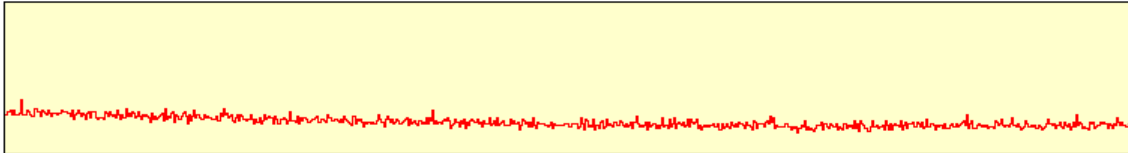
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Symphony A Tim System SaTs : Body Coil with Scale Factor = 10

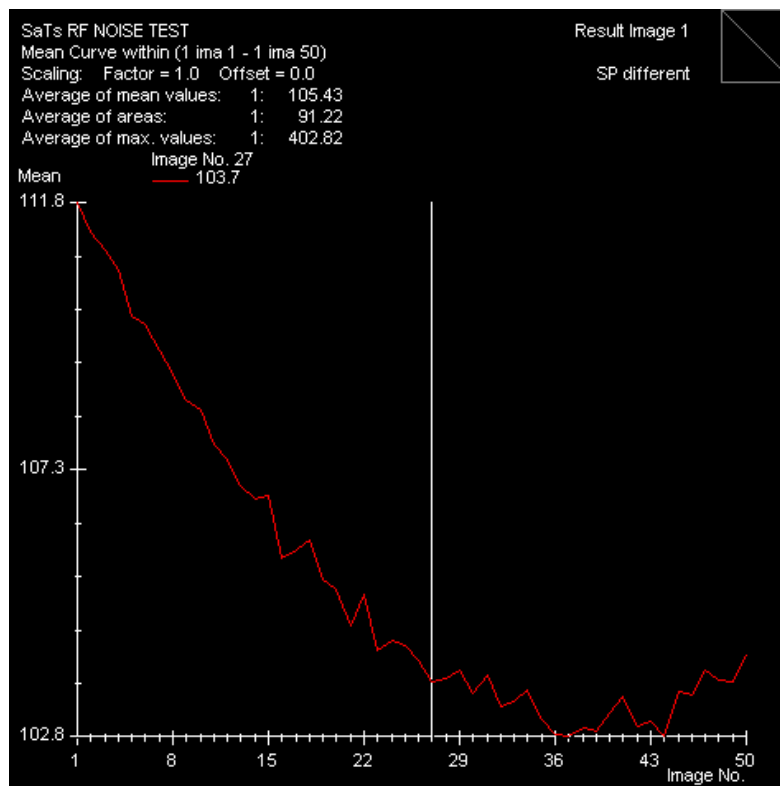
Frequency 63.306020 - 63.806020 Mhz

Center Frequency 63.556020 Mhz



Maximum Column Mean Value: 126

Minimum Column Mean Value: 91



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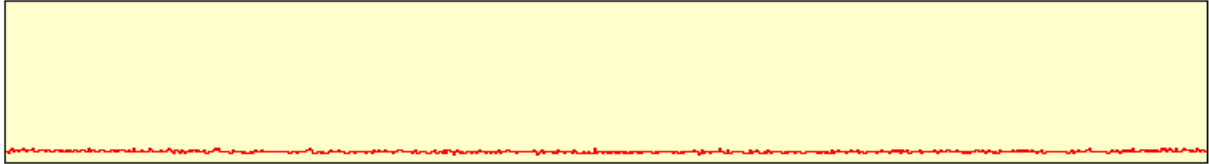
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Verio : Body Coil with Scale Factor = 10

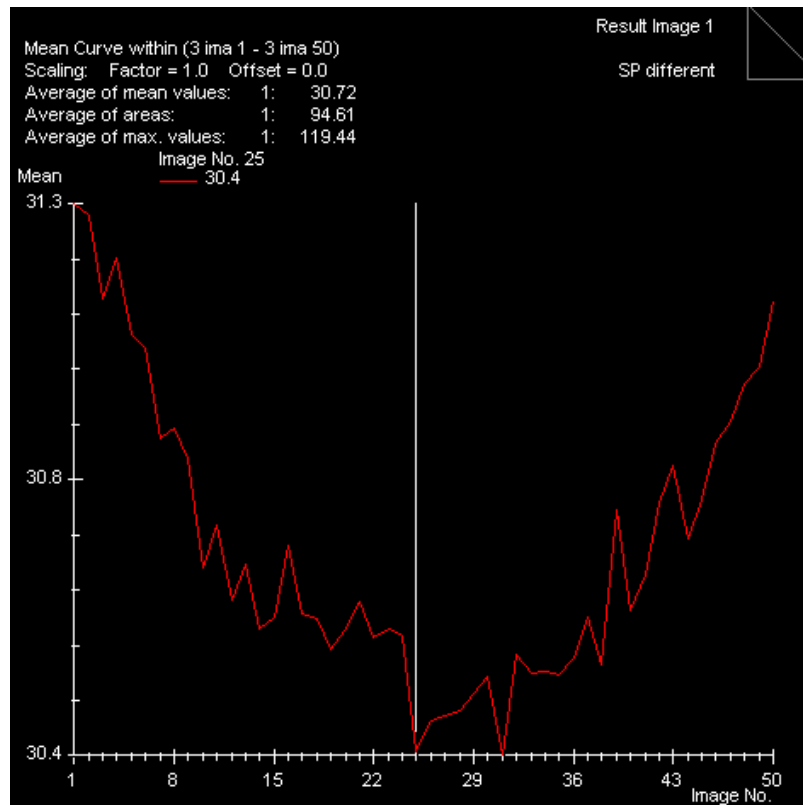
Frequency 122.939592 - 123.439592 Mhz

Center Frequency 123.189592 Mhz



Maximum Column Mean Value: 34

Minimum Column Mean Value: 26



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Frequently Asked Questions

Question: I cannot get the scan to run when I turn off power to the RF room?

Answer:

On VB13 and higher software the software employs scan on center, during the scan start the MPCU communicates with MDSD about table positioning and this cannot be simulated or masked out. Workaround is to set number of measurements to 2, start the scan then power off the RF room. Ignore the first measurement and only review the second measurement.

Question: How can I calculate the frequency of the interference?

Answer:

For systems with software version less than VB13 and VA30 use the following formula:
 $(\text{Center Frequency} - 250000) + (\text{Image number} \times 10000) + (\text{Column number} \times (10000/255))$

For system with software version VB13 and higher and VA30 use the following formula:
 $(\text{Center Frequency} - 250000) - (\text{Image number} \times 10000) + (\text{Column number} \times (10000/255))$

For Concerto and C! With software version less than VA30 use the following formula:
 $(\text{Center Frequency} - 250000) + (\text{Image number} \times 10000) + (255 - \text{Column number} \times (10000/255))$

For system with software version VA30 use the following formula:
 $(\text{Center Frequency} - 250000) - (\text{Image number} \times 10000) + (255 - \text{Column number} \times (10000/255))$

Question: My profiles are reverse of the examples in this document or other systems?

Answer:

At the introduction of VA30 and VB13 they changed which images numbers corresponded to the low and high frequency range. This resulted in the older software the profile for the frequency range was left to right for low to high and the new software left to right for high to low frequency range.
